IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
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- Conclusion
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Executive Summary

Historical Falcon 9 launch data were analyzed to identify factors affecting landing success. Payload mass, launch site, orbit type, and launch year were found to be important predictors. Multiple classification models achieved approximately 83% accuracy, demonstrating that landing outcomes can be predicted effectively from historical mission data.

Introduction

Project Background and Context

SpaceX has significantly reduced the cost of space launches by reusing the Falcon 9 first stage booster. While a Falcon 9 launch costs approximately $62 million, similar launches with expendable rockets can be much more expensive. The ability to predict whether the first stage will successfully land is important because it directly impacts mission costs and operational planning. This project analyzes historical Falcon 9 launch data to identify the factors that influence landing success and to build machine learning models capable of predicting landing outcomes.

Accurate prediction of landing success can support mission planning, risk assessment, and cost optimization for future launch operations.

Introduction

Problems to Be Solved

- Which factors have the greatest impact on Falcon 9 first-stage landing success?
- How do payload mass, launch site, orbit type, and launch year affect landing outcomes?
- Are there observable trends in landing success rates over time?
- Can machine learning models accurately predict whether a Falcon 9 first stage will successfully land?
- Which classification model provides the best predictive performance?

Section 1

Methodology



Methodology

Executive Summary

- Launch data were collected from public SpaceX sources using API requests and web scraping techniques. The datasets included information such as payload mass, launch site, orbit type, launch date, and landing outcomes. The collected data were consolidated into a structured dataset for further analysis.
- The collected data were cleaned and prepared for analysis by handling missing values, removing inconsistencies, and standardizing variable formats. Categorical variables were transformed into suitable formats for machine learning models. The resulting dataset provided a reliable foundation for exploratory and predictive analysis.

Methodology

Executive Summary

- Exploratory data analysis was performed using both visualization techniques and SQL queries. Scatter plots, bar charts, and trend analyses were used to examine relationships between payload mass, launch sites, orbit types, and landing outcomes. SQL analysis was used to retrieve key statistics and identify patterns within the launch data.Key Findings:Landing success rates improved over time.Certain launch sites achieved higher success rates.Orbit type significantly influenced landing outcomes.Payload mass showed a measurable relationship with landing success.

Methodology

Executive Summary

- Interactive Visual AnalyticsInteractive visual analytics were developed using Folium and Plotly Dash. Folium maps were used to visualize launch site locations and their surrounding geographical features. Plotly Dash dashboards allowed users to dynamically explore launch statistics, payload distributions, and success rates across different mission characteristics.
- Predictive Analysis Using Classification Models

Machine learning classification models were developed to predict Falcon 9 first-stage landing success. Feature engineering and data standardization were applied before model training. Multiple algorithms were compared to determine the most effective approach for classification.Models evaluated: Logistic Regression, Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbors (KNN)

Methodology

Executive Summary

Building, Tuning and Evaluating Models

The dataset was split into training and testing subsets. Hyperparameter tuning was performed using GridSearchCV to identify optimal model configurations. Model performance was evaluated using accuracy scores and confusion matrices. The best-performing models achieved an accuracy of approximately 83%, indicating strong predictive capability for Falcon 9 landing outcomes.Evaluation Metrics:

- AccuracyConfusion
- Matrix
- Cross Validation
- Hyperparameter Optimization

Data Collection

- Data was collected from the SpaceX REST API and Wikipedia launch records. The data was extracted, combined, cleaned, and prepared for analysis.

Data Collection – SpaceX API

Data Collection using SpaceX REST API

- Connected to the SpaceX REST API
- Retrieved Falcon 9 launch records
- Collected rocket, payload, launch site, and core data

- https://github.com/DsUgurPolat/010-Angewandte-Datenwissenschaft-Capstone/blob/main/Modul%201/004%20jupyter-labs-spacex-data-collection-api-v2.ipynb

- SpaceX REST API
- Retrieve Launch Data
- Extract Required Fields(Flight No., Payload,Launch Site, Outcome)
- Merge Related Data(Rocket, Payload, Core)
- Create DataFrame
- Export for Analysis

Data Collection - Scraping

- Launch data was collected from Wikipedia using web scraping techniques. HTML tables were extracted, parsed, cleaned, and converted into a structured dataset for analysis.
- https://github.com/DsUgurPolat/010-Angewandte-Datenwissenschaft-Capstone/blob/main/Modul%201/005%20jupyter-labs-webscraping.ipynb

- Wikipedia Page
- Retrieve HTML Content
- Extract Launch Tables
- Parse Required Fields
- Clean & Transform Data
- Create DataFrame
- Store Dataset

Data Wrangling

- Cleaned the collected data, handled missing values, selected relevant Falcon 9 launches, and prepared the dataset for exploratory analysis and machine learning
- Raw Data-Remove Missing Values-Filter Falcon 9 Launches-Create Landing Outcome Label-Select Relevant Features-Clean Dataset-Ready for Analysis
- https://github.com/DsUgurPolat/010-Angewandte-Datenwissenschaft-Capstone/blob/main/Modul%201/007%20labs-jupyter-spacex-Data%20wrangling-v2.ipynb

EDA with Data Visualization

Charts Used

- Flight Number vs. Launch Site (Scatter Plot)
- Payload Mass vs. Launch Site (Scatter Plot)
- Success Rate by Orbit Type (Bar Chart)
- Payload Mass vs. Orbit Type (Scatter Plot)
- Yearly Launch Success Trend (Line Chart)

EDA with Data Visualization

Purpose

- Explore relationships between launch variables
- Identify factors affecting landing success
- Analyze trends and patterns in Falcon 9 launches
- Support feature selection for predictive modeling

Data visualization was used to identify patterns, trends, and correlations between launch site, payload mass, orbit type, and mission success.

- https://github.com/DsUgurPolat/010-Angewandte-Datenwissenschaft-Capstone/blob/main/Modul%202/003%20jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

- **SQL queries were used to explore launch records, payload statistics, landing outcomes, mission success rates, and booster performance.**

- https://github.com/DsUgurPolat/010-Angewandte-Datenwissenschaft-Capstone/blob/main/Modul%202/002%20jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- Folium maps were used to visualize launch sites, mission outcomes, and nearby geographical features. These map objects helped identify location patterns and understand the operational environment of each launch site.

Map Objects Created

- Markers for SpaceX launch sites
- Colored markers for launch outcomes (success/failure)
- Circle markers to highlight launch locations
- Lines connecting launch sites to nearby infrastructure
- Distance markers showing proximity to roads, railways, coastlines, and cities

Build an Interactive Map with Folium

Why These Objects Were Added

- Markers identify launch site locations.
- Colored markers visualize mission outcomes.
- Circle markers improve location visibility.
- Lines show relationships between launch sites and nearby features.
- Distance markers help analyze factors that may influence launch operations.

- https://github.com/DsUgurPolat/010-Angewandte-Datenwissenschaft-Capstone/blob/main/Modul%203/002_lab-jupyter-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

A Plotly Dash dashboard was developed to visualize launch success rates and the impact of payload mass on mission outcomes. Interactive filters enabled dynamic exploration of launch data.

Plots/Graphs Added

- Pie Chart: Launch success count by launch site
- Pie Chart: Success rate for a selected launch site
- Scatter Plot: Payload Mass vs. Launch Outcome
- Payload Range Slider: Interactive payload filtering
- Launch Site Dropdown: Interactive site selection

Build a Dashboard with Plotly Dash

Why These Interactions Were Added

- Compare launch performance across sites
- Analyze launch success rates
- Explore the relationship between payload mass and landing outcome
- Allow users to filter and investigate specific launch conditions
- Improve data exploration through interactive visualization

- https://github.com/DsUgurPolat/010-Angewandte-Datenwissenschaft-Capstone/blob/main/Modul%203/003_Build_a_Dashboard.ipynb

Predictive Analysis (Classification)

Best performing classification model Short Summary

- Selected relevant features and standardized data.
- Split the dataset into training and testing sets.
- Built Logistic Regression, SVM, Decision Tree, and KNN models.
- Tuned model parameters using GridSearchCV.
- Compared classification accuracy on test data.
- Selected the model with the highest accuracy as the final classifier.

Predictive Analysis (Classification)

Model Development Flowchart

- Prepared Dataset
- Feature Selection
- Data Standardization
- Train-Test Split
- Build Classification Models
- (Logistic Regression,
- SVM, Decision Tree,
- KNN)
- Hyperparameter Tuning
- (GridSearchCV)
- Evaluate Accuracy
- Select Best Model

- https://github.com/DsUgurPolat/010-Angewandte-Datenwissenschaft-Capstone/blob/main/Modul%204/002_SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

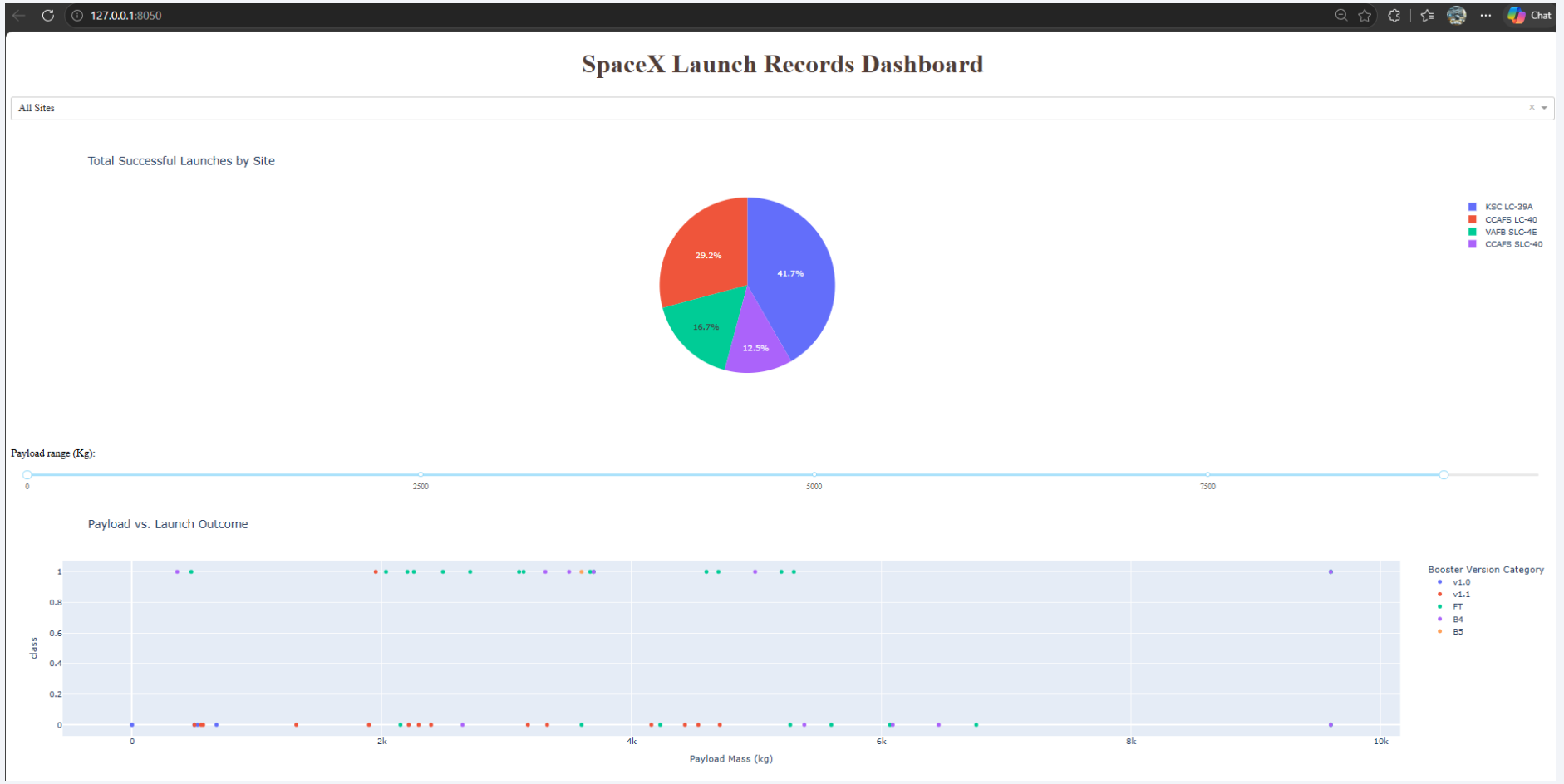
Results

Exploratory Data Analysis Results

- Launch success rate improved over time.
- Payload mass and orbit type influenced landing success.
- Some launch sites achieved higher success rates than others.
- Successful landings became more frequent in later missions.

Results

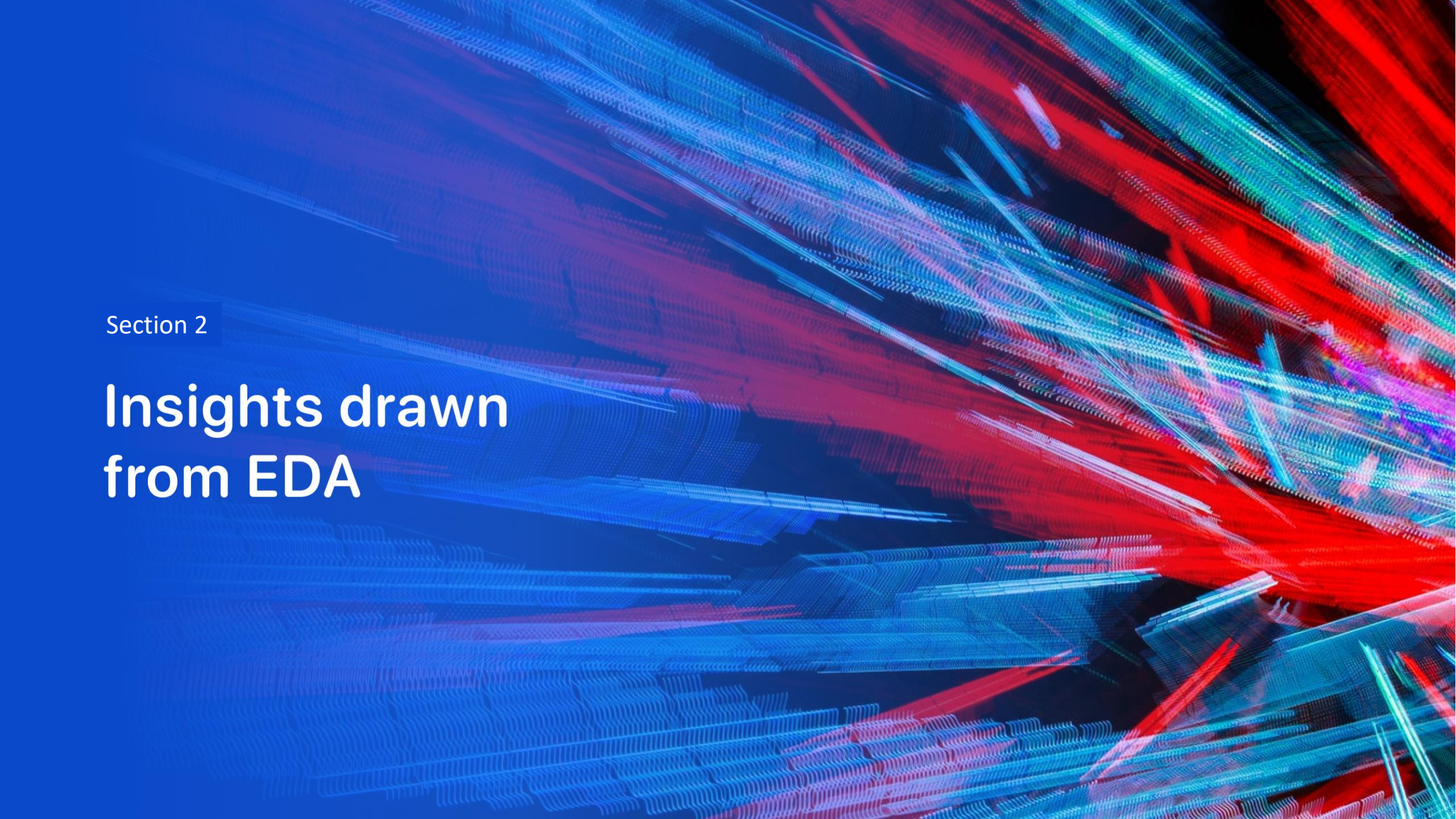
Interactive analytics demo in screenshots



Results

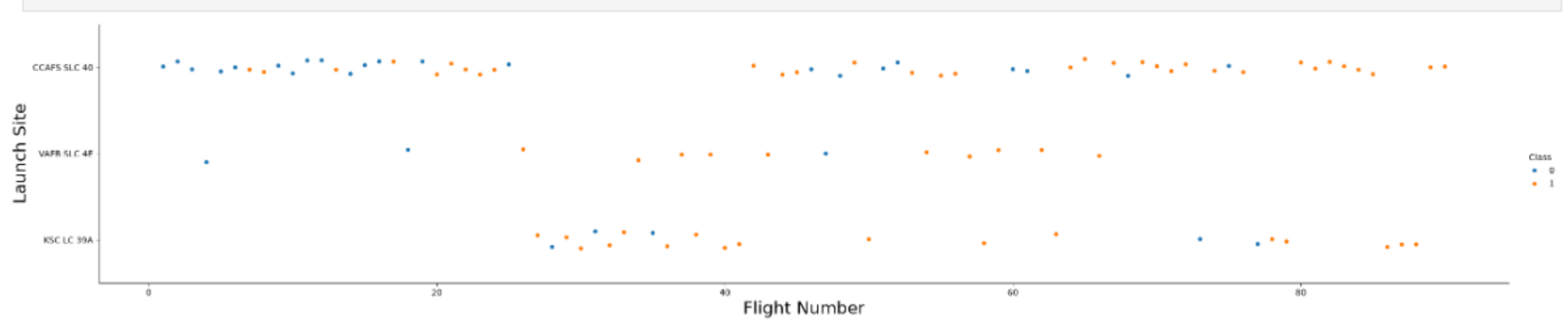
Predictive Analysis Results

- Multiple classification models were trained and evaluated.
- Hyperparameter tuning improved model performance.
- The best model achieved the highest prediction accuracy.
- The final model successfully predicted Falcon 9 landing outcomes.


Section 2
Insights drawn from EDA

Flight Number vs. Launch Site

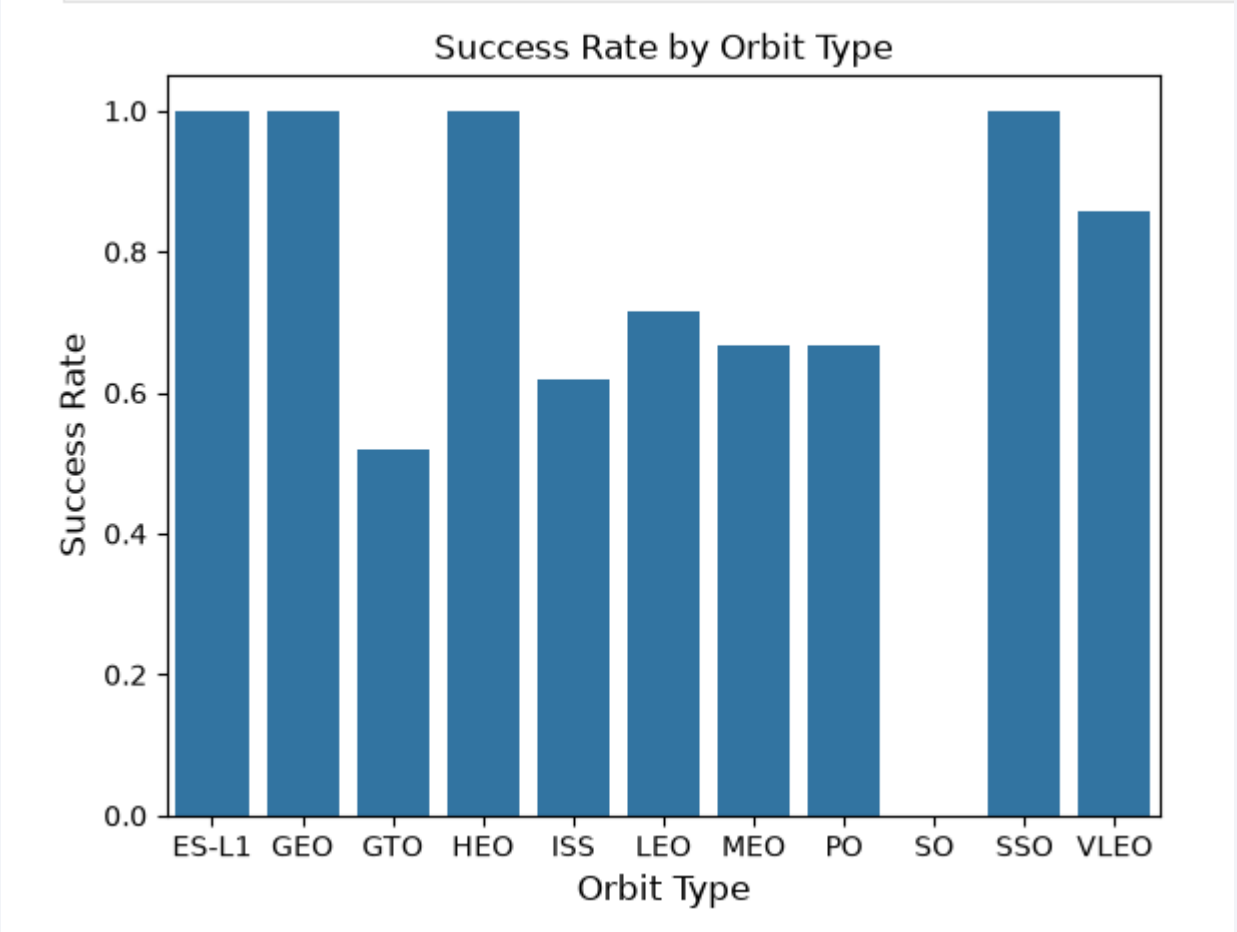
- Scatter plot of Flight Number vs. Launch Site



The scatter plot shows that launch activity expanded over time and became more successful as SpaceX gained operational experience. Florida launch sites handled most missions, while VAFB supported specialized orbit launches.

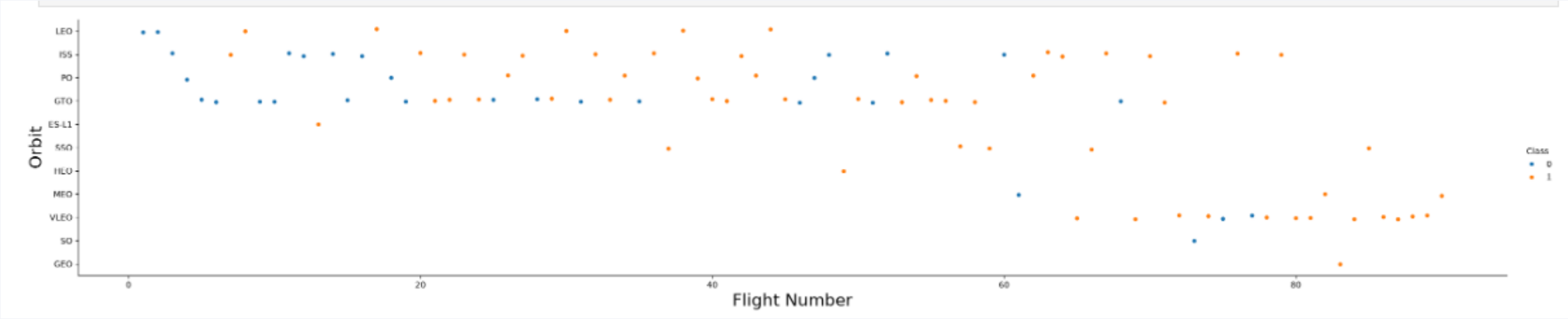
Success Rate vs. Orbit Type

The bar chart shows that mission success rates vary across orbit types. LEO missions achieved consistently high success rates, while GTO missions were more challenging and exhibited a lower success rate. Overall, Falcon 9 demonstrated strong performance across most orbit categories.



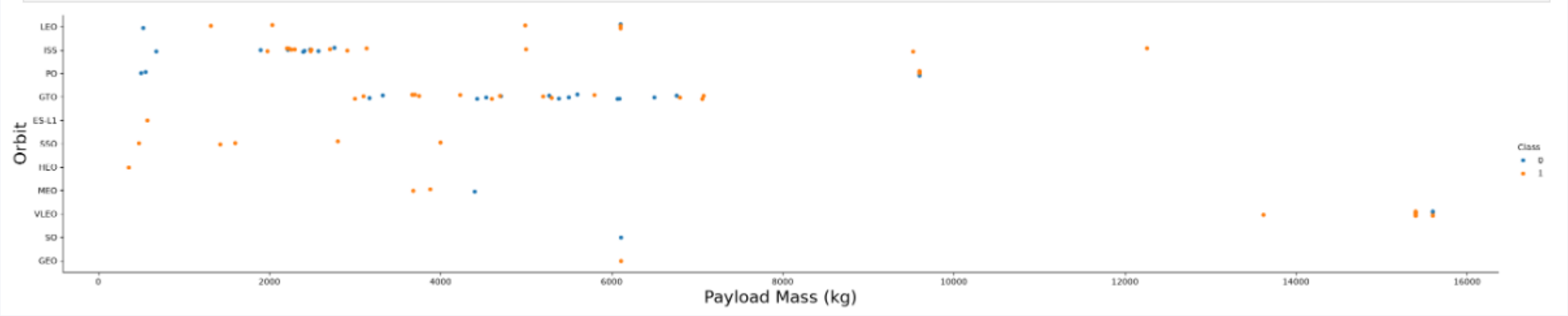
Flight Number vs. Orbit Type

- The orbit type is an important factor affecting mission success. Higher success rates were observed for several orbit categories, while GTO missions presented greater operational challenges.



Payload vs. Orbit Type

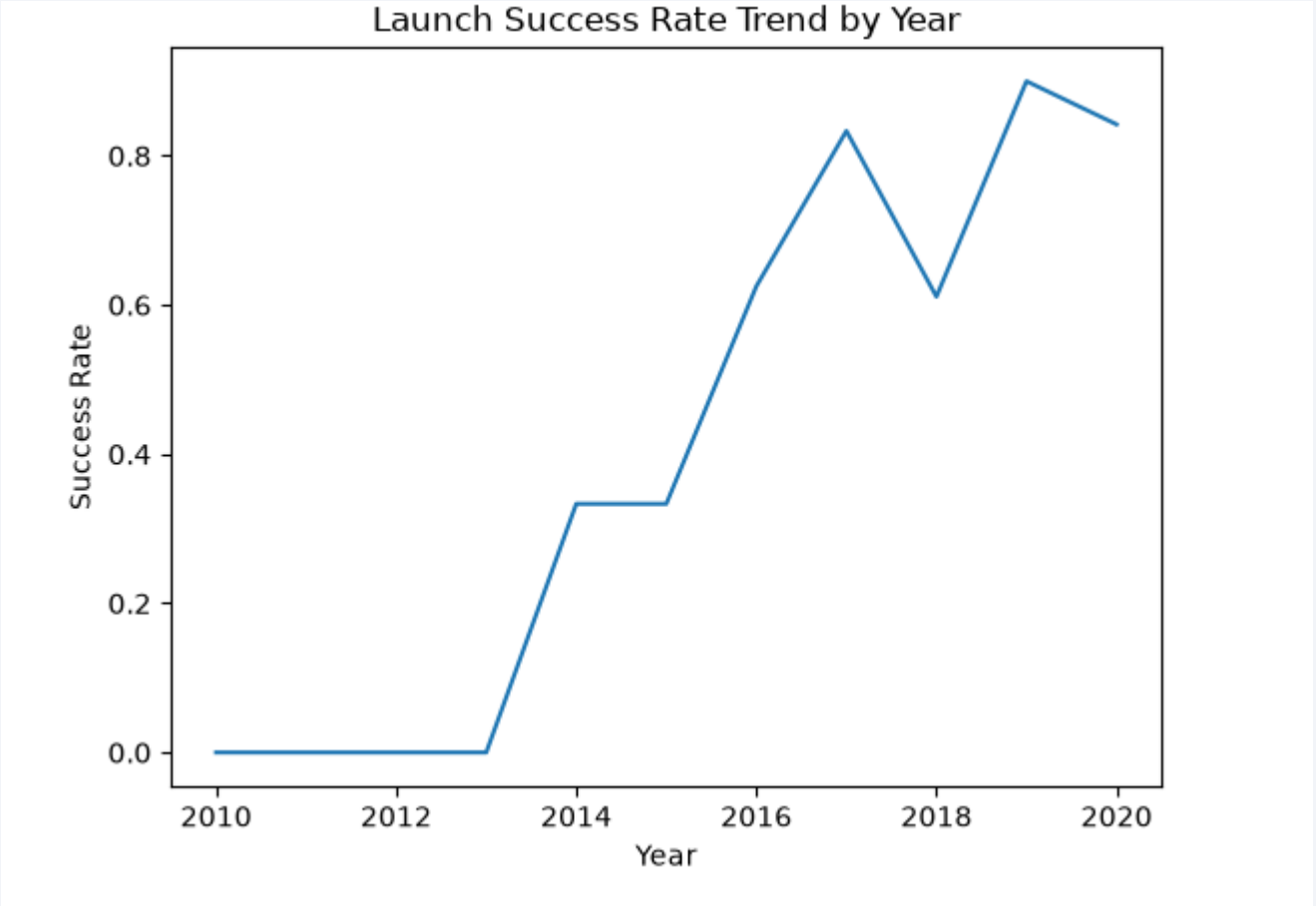
The scatter plot shows the relationship between payload mass and orbit type. Different orbit categories require different payload capacities, which influences mission characteristics. While heavier payloads can increase mission complexity, SpaceX achieved successful landings across multiple orbit types and payload ranges.



Launch Success Yearly Trend

Explanation

The line chart illustrates the yearly average launch success rate. The upward trend indicates continuous improvements in Falcon 9 performance, landing procedures, and operational efficiency. Over time, SpaceX achieved more reliable launch outcomes and booster recoveries.



All Launch Site Names

- **Query Result**
  - CCAFS LC-40
  - CCAFS SLC-40
  - KSC LC-39A
  - VAFB SLC-4E
- **Short Explanation**

The SQL query was used to identify all unique SpaceX launch sites in the dataset. Most launches were conducted from Florida launch facilities (CCAFS and KSC), while VAFB SLC-4E supported missions to polar and sun-synchronous orbits.

Launch Site Names Begin with 'CCA'

	Date	Time_UTC	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	2010-06-04	0 days 18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0.0	LEO	SpaceX	Success	Failure (parachute)\r
1	2010-12-08	0 days 15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0.0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)\r
2	2012-05-22	0 days 07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525.0	LEO (ISS)	NASA (COTS)	Success	No attempt\r
3	2012-10-08	0 days 00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500.0	LEO (ISS)	NASA (CRS)	Success	No attempt\r
4	2013-03-01	0 days 15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677.0	LEO (ISS)	NASA (CRS)	Success	No attempt\r

The SQL query filtered launch records where the launch site name begins with "CCA". The results show that Cape Canaveral Air Force Station (CCAFS) was one of SpaceX's primary launch locations and hosted numerous Falcon 9 missions.

Total Payload Mass

- **Query Result**

Total Payload Mass Carried by NASA Boosters: 45,596 kg

- **Short Explanation**

The SQL query calculated the total payload mass transported on NASA missions. The result shows that Falcon 9 boosters carried a combined payload of 45,596 kg for NASA-related launches, demonstrating SpaceX's significant contribution to NASA space missions.

Average Payload Mass by F9 v1.1

- **Query Result**

Average Payload Mass Carried by F9 v1.1: 2,928.4 kg

- **Short Explanation**

The SQL query calculated the average payload mass carried by the Falcon 9 v1.1 booster version. The result shows that F9 v1.1 missions transported an average payload of 2,928.4 kg, providing insight into the launch capabilities of this booster configuration.

First Successful Ground Landing Date

- **Query Result**

First Successful Ground Landing Date: 2015-12-22

- **Short Explanation**

The SQL query identified the first mission that achieved a successful landing on a ground pad. This milestone demonstrated SpaceX's progress in developing reusable rocket technology and paved the way for routine booster recovery.

Successful Drone Ship Landing with Payload between 4000 and 6000

- **Query Result**
  - F9 FT B1022
  - F9 FT B1026
  - F9 FT B1021.2
  - F9 FT B1031.2
- **Short Explanation**

The SQL query identified booster versions that successfully landed on a drone ship while carrying payloads between 4,000 kg and 6,000 kg. The results demonstrate Falcon 9's ability to recover boosters successfully even during medium-payload missions.

Total Number of Successful and Failure Mission Outcomes

Query Result

- Successful Missions: 100
- Failed Missions: 1

Short Explanation

The SQL query counted all mission outcomes in the dataset. The results show that successful missions significantly outnumbered failed missions, reflecting the continuous improvement and reliability of the Falcon 9 launch system.

Boosters Carried Maximum Payload

Query Result

- F9 B5 B1048.4
- F9 B5 B1049.4
- F9 B5 B1051.3
- F9 B5 B1056.4
- F9 B5 B1048.5
- F9 B5 B1051.4
- F9 B5 B1049.5
- F9 B5 B1060.2
- F9 B5 B1058.3
- F9 B5 B1051.6
- F9 B5 B1060.3
- F9 B5 B1049.7

Short Explanation

The SQL query identified the Falcon 9 Block 5 boosters that carried the maximum payload mass recorded in the dataset. Multiple booster cores achieved this payload capacity, demonstrating the high performance and operational reliability of the Falcon 9 Block 5 generation.

2015 Launch Records

Query Result			
Month	Landing Outcome	Booster Version	Launch Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Short Explanation

The SQL query identified all failed drone ship landing attempts in 2015. Two Falcon 9 v1.1 missions failed to land successfully on the drone ship, both launched from CCAFS LC-40. These early failures provided valuable experience that helped SpaceX improve its booster recovery technology in later missions.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

	Landing_Outcome	Count_Landing_Outcome
0	No attempt\r	10
1	Failure (drone ship)\r	5
2	Success (drone ship)\r	5
3	Controlled (ocean)\r	3
4	Success (ground pad)\r	3
5	Failure (parachute)\r	2
6	Uncontrolled (ocean)\r	2
7	Precluded (drone ship)\r	1

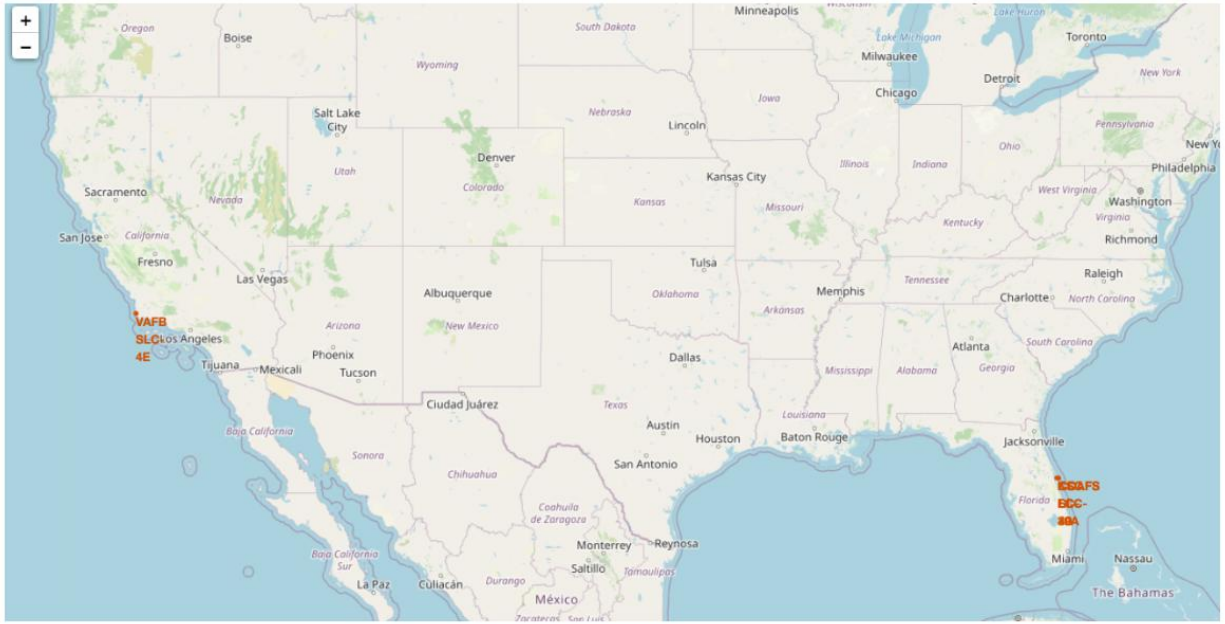
Short Explanation

The landing outcomes were ranked by frequency between 2010 and 2017. "No attempt" was the most common outcome with 10 missions. Both successful and failed drone ship landings occurred 5 times, reflecting SpaceX's early efforts to develop reusable booster technology. Successful ground pad landings and controlled ocean landings became more frequent as recovery capabilities improved.


Section 3
Launch Sites
Proximities Analysis

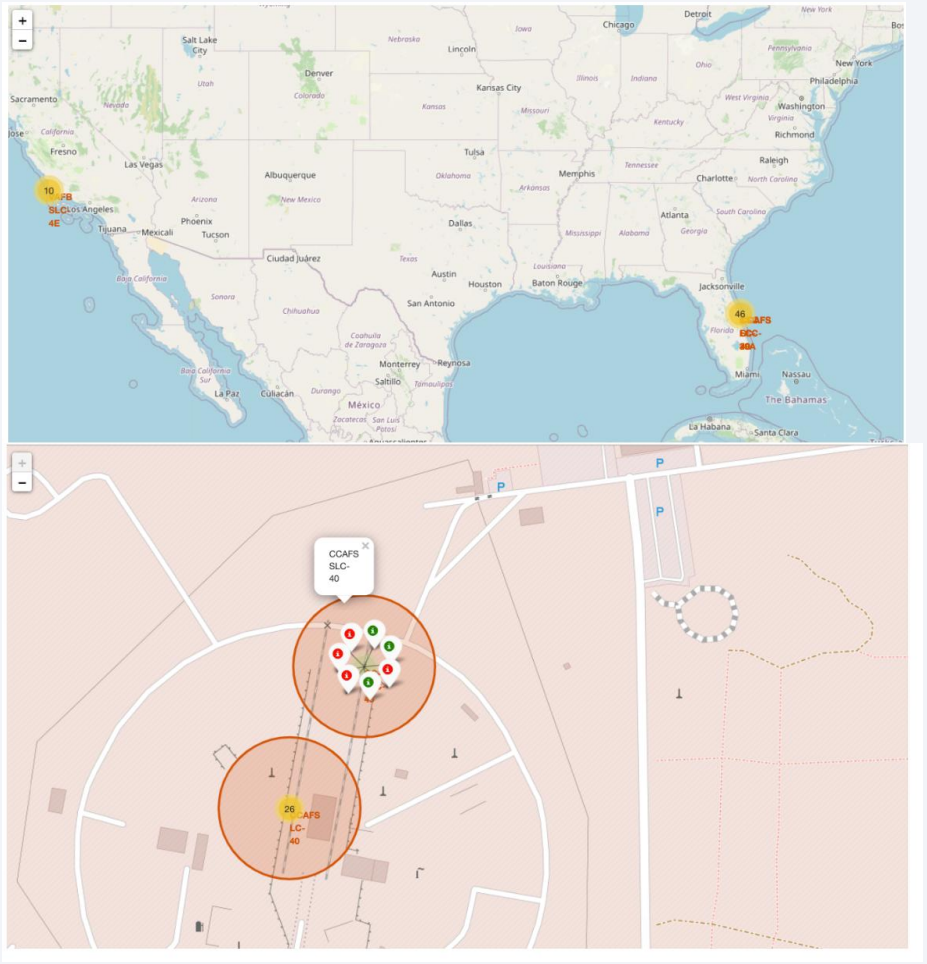
Global Distribution of SpaceX Launch Sites

The Folium map visualizes all SpaceX launch sites using location markers. The map shows that SpaceX primarily operates from Florida launch facilities, while VAFB SLC-4E supports specialized missions from the west coast. The geographic distribution of launch sites enables flexibility for different launch trajectories and mission objectives.



Launch Outcomes Visualization on Folium Map

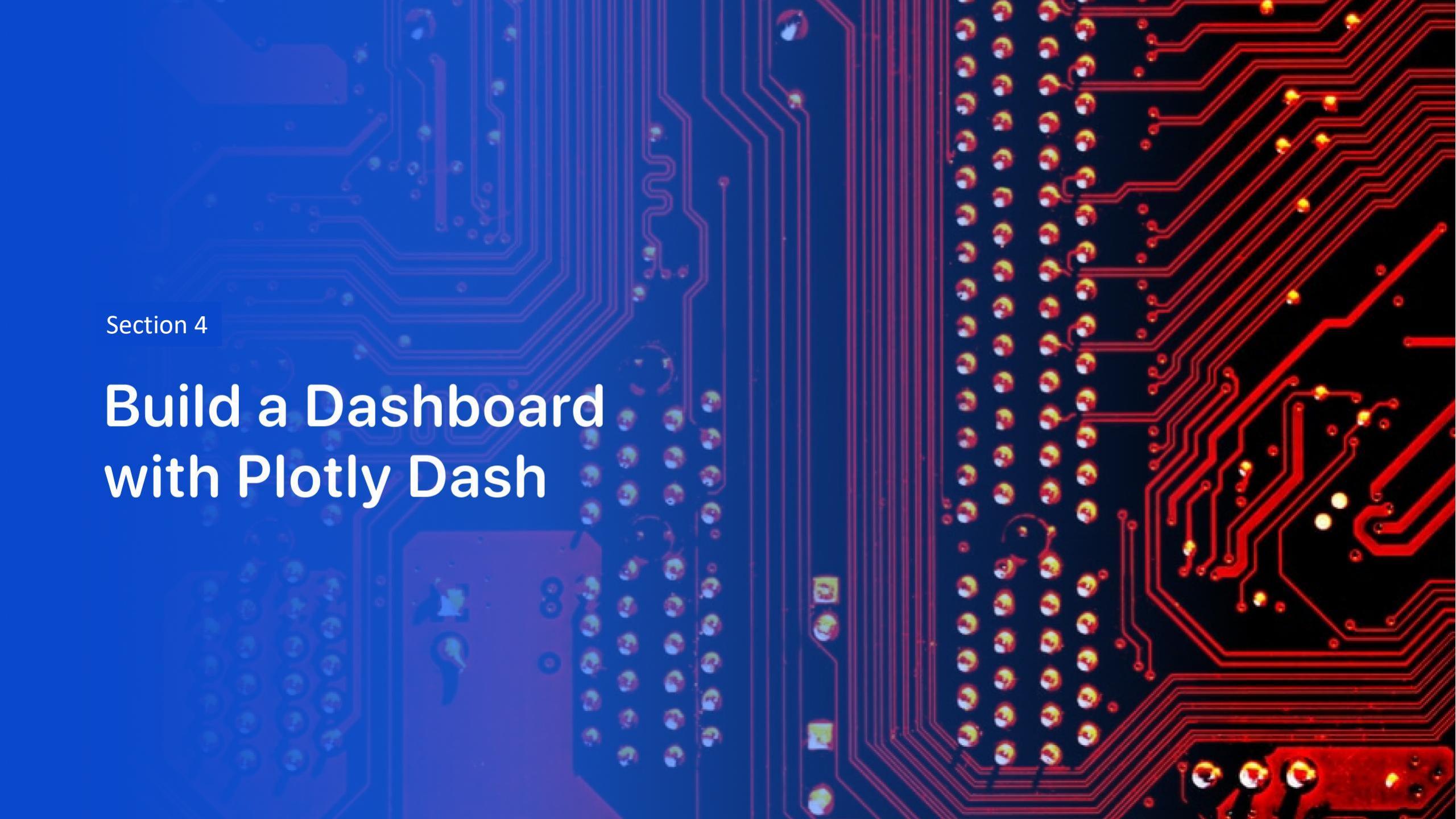
The Folium map uses color-coded markers to visualize launch outcomes. Green markers indicate successful landings, while red markers indicate failures. The distribution of colors shows that successful recoveries became increasingly common as SpaceX improved its booster landing technology.



Launch Site Proximity Analysis

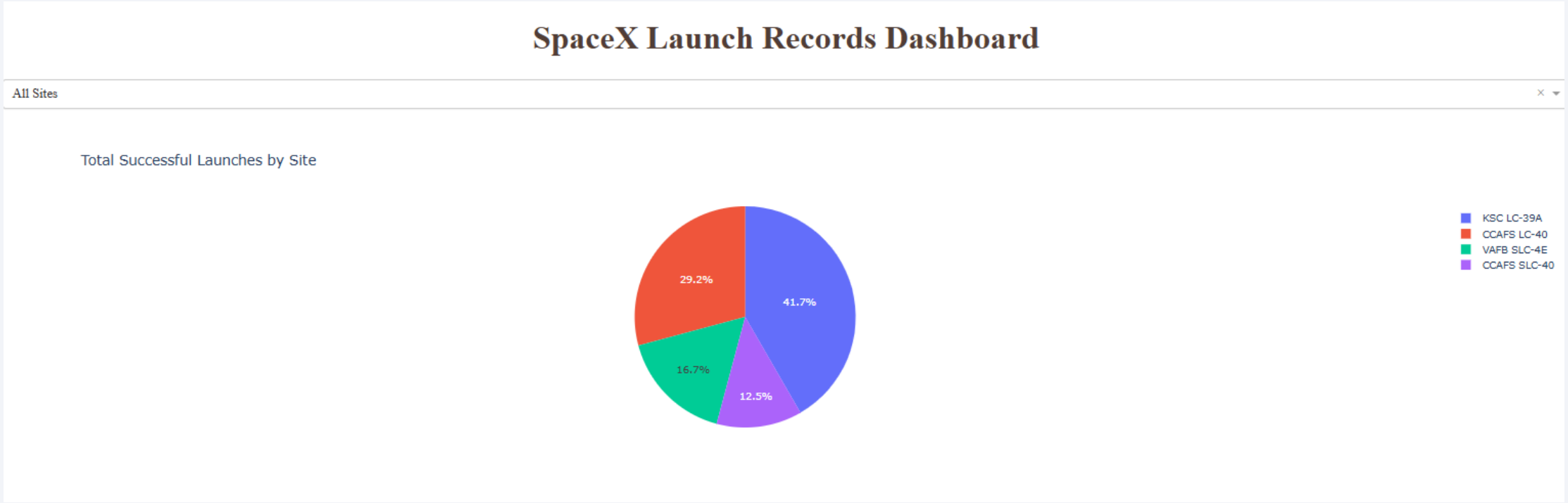
The Folium map visualizes the selected launch site's proximity to nearby infrastructure and geographical features. Distance measurements to the coastline, highways, and railways help evaluate the operational advantages of the launch location. The analysis shows that SpaceX launch sites are strategically positioned to balance safety, accessibility, and mission requirements




Section 4
Build a Dashboard with Plotly Dash

Launch Success Count by Launch Site

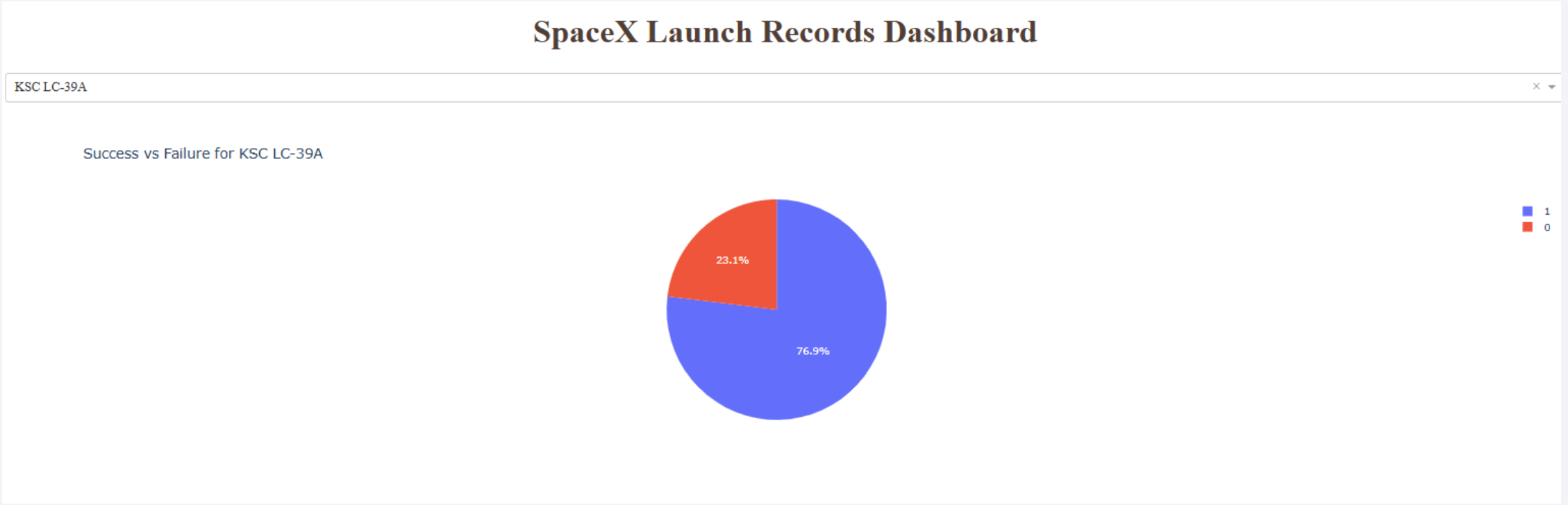
The pie chart displays the distribution of successful launches across all SpaceX launch sites. The largest portions correspond to the sites with the highest number of successful missions, demonstrating that Florida-based launch facilities handled most Falcon 9 operations and contributed significantly to overall mission success.



Launch Site with the Highest Success Ratio

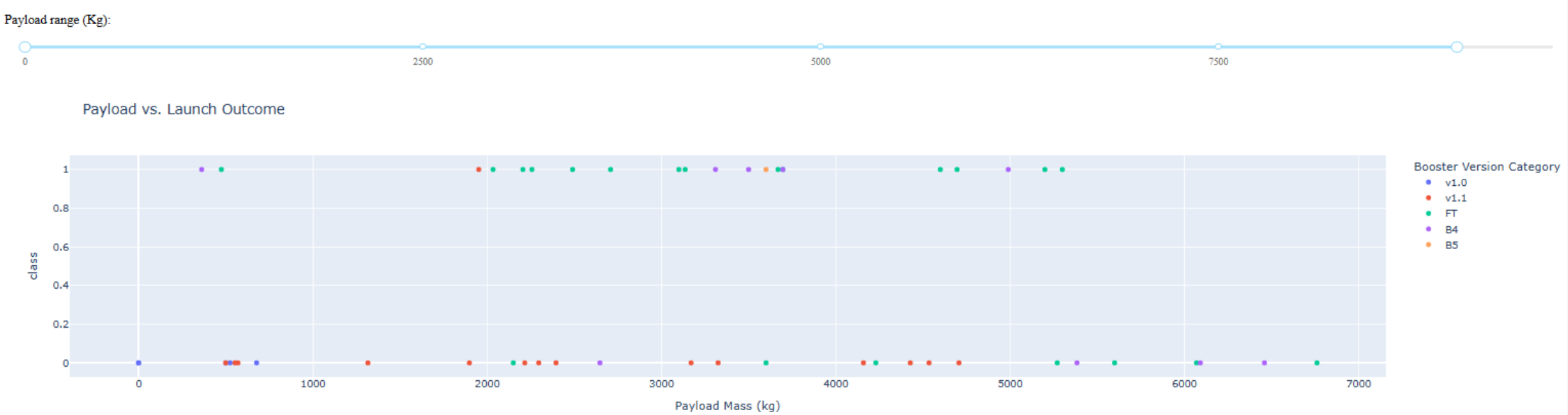
The pie chart shows the distribution of successful and unsuccessful launches for the launch site with the highest success ratio. The large proportion of successful missions demonstrates the strong performance and reliability achieved at this location.

Key Takeaway: This launch site recorded the highest launch success ratio, making it one of the most reliable SpaceX launch facilities in the dataset.



Payload Mass vs. Launch Outcome Analysis

Short Explanation: The scatter plot illustrates the relationship between payload mass and launch outcome. By adjusting the payload range slider, different mission groups can be explored. The results show that successful landings occurred across a wide range of payload masses, indicating that Falcon 9 boosters can reliably recover even during demanding missions.Key TakeawayFalcon 9 achieved high landing success rates across various payload ranges, with newer booster versions delivering the most consistent performance.



Section 5

Predictive Analysis (Classification)



Classification Accuracy

ModelAccuracy:

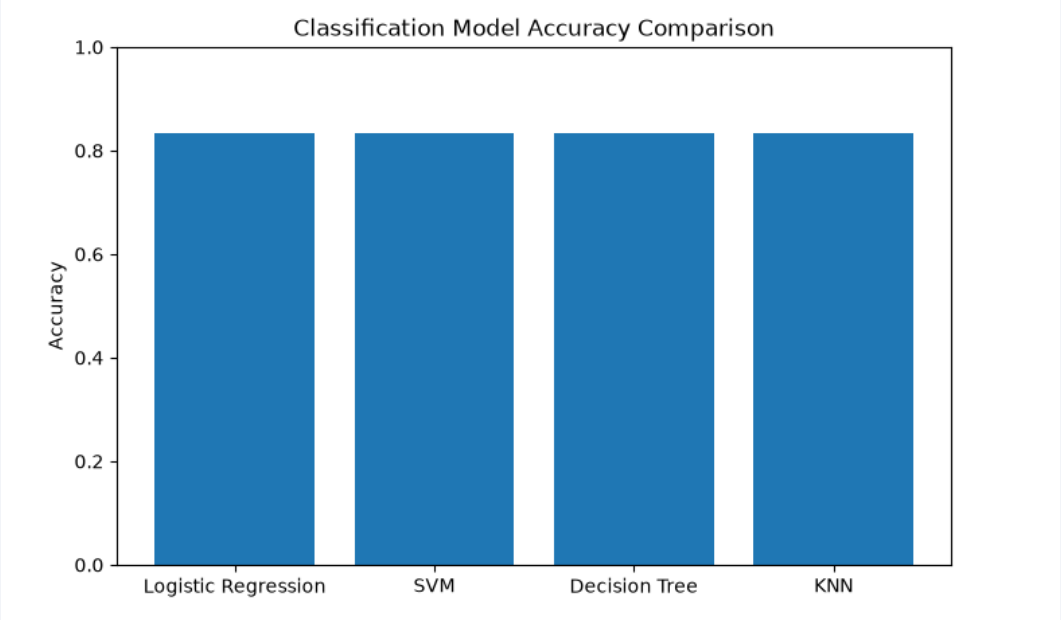
LogisticRegression: % 83.3

SVM: % 83.3

Decision Tree : % 83.3

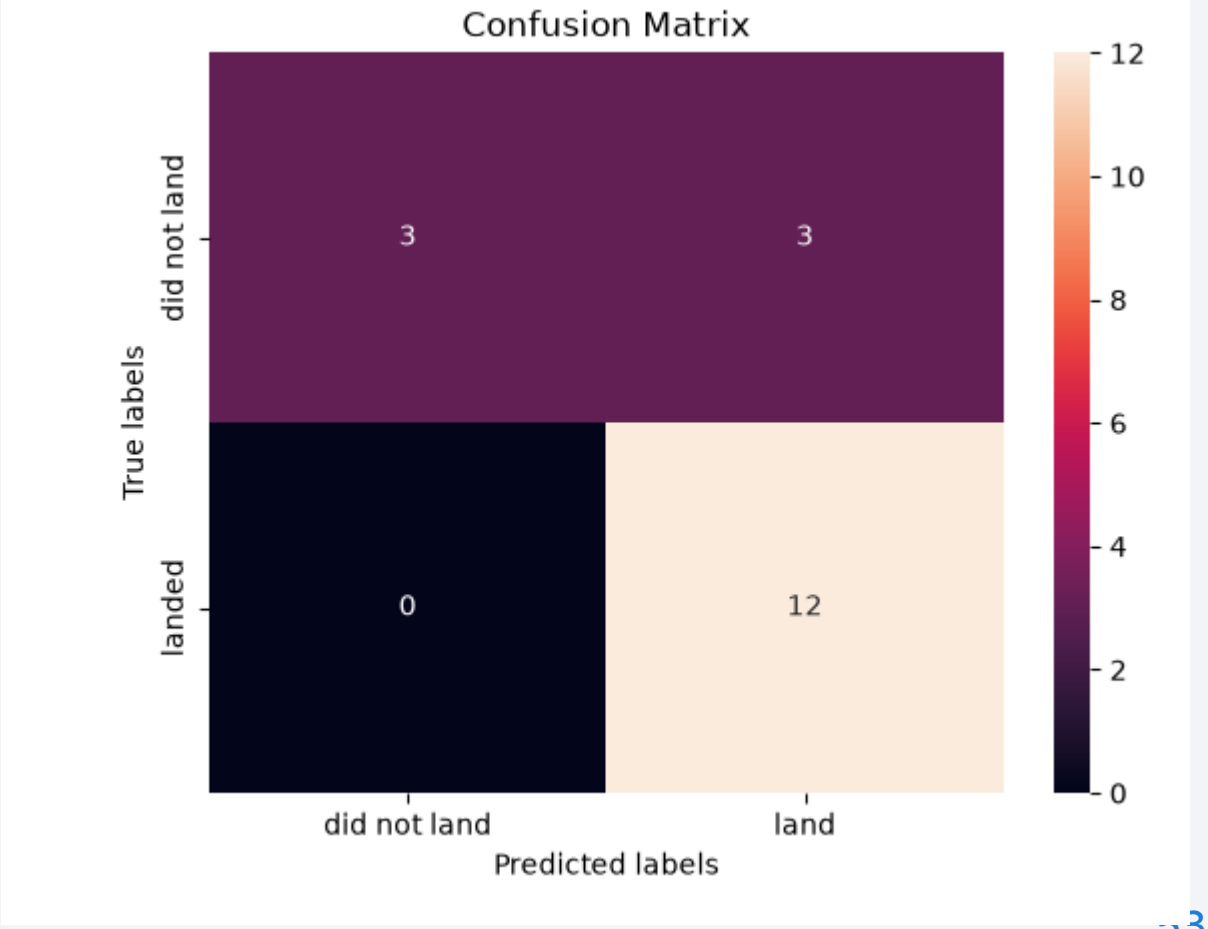
KNN : % 83.3

All models achieved similar performance. The results indicate that historical launch characteristics contain sufficient information to accurately predict Falcon 9 landing outcomes.



Confusion Matrix

The confusion matrix of KNN shows that the model correctly classified 12 successful landings and 3 unsuccessful landings. No successful landings were missed. However, 3 unsuccessful landings were incorrectly predicted as successful landings. Overall, the model demonstrates strong performance in predicting Falcon 9 landing outcomes.



Conclusions

- Data from the SpaceX API and Wikipedia was successfully collected, cleaned, and prepared for analysis.
- Exploratory Data Analysis identified launch site, orbit type, and payload mass as important factors influencing landing success.
- Interactive Folium maps and Plotly Dash dashboards provided valuable insights into launch performance and geographic patterns.
- Falcon 9 mission success rates improved significantly over time, demonstrating advancements in reusable rocket technology.
- Multiple machine learning models were developed and evaluated for landing outcome prediction.
- Logistic Regression, SVM, Decision Tree, and KNN all achieved an accuracy of 83.33% on the test dataset.
- The predictive models demonstrated that Falcon 9 landing outcomes can be forecasted effectively using historical launch data.
- Overall, the project shows how data science techniques can support decision-making and performance analysis in the aerospace industry.

Conclusions

- Orbit type, payload mass, launch site, and launch year significantly influenced landing success.
- Landing success rates improved steadily over time. The success rate increased significantly after the introduction of booster recovery improvements, suggesting that technological advancements played a major role in landing reliability.
- Predictive analytics can support mission planning, operational decision-making, and cost optimization.

Appendix

Supporting Materials

- Python Code
- SQL Queries
- Dashboard Screenshots
- Folium Maps
- Model Results
- Confusion Matrix
- Dataset Samples
- GitHub Repository Link

Thank you!

